

Maverick Luke

+1 (437) 981 2039 | maveluk.cy@gmail.com | linkedin.com/in/maveluke | github.com/Maveluke

EDUCATION

University of Toronto

HBSc, Specialist in Data Science and Major in Computer Science

- CGPA: 3.79, **Dean's List** 2024 and 2025

Toronto, Ontario

Expected June 2026

EXPERIENCE

Full-Stack Developer → Tech Lead

Aug 2025 - Present

Ontario Association of Physics Teachers (OAPT), University of Toronto

Toronto, Ontario

- Promoted to **Tech Lead** after 5 months as Full-Stack Developer, owning end-to-end architecture and development of a contest platform designed for **660,000+ students** across **883+ Ontario schools**
- Architected **scalable** full-stack web application using **React**, **Node.js/Express.js**, **PostgreSQL**, and **Prisma ORM**, designing a normalized database schema and **service-oriented RESTful API** with **60+** endpoints following MVC principles
- Built secure authentication with **JWT tokens**, **bcrypt** password hashing, and **role-based access control (RBAC)**; used **PostgreSQL transactions** to ensure data integrity across multi-step operations
- Designed and implemented the **contest editing feature** end-to-end, integrating **TipTap/ProseMirror** rich-text editing with **LaTeX** math rendering, drag-and-drop question ordering, and **PDF generation** via **Puppeteer**
- Lead a team of **8 developers** alongside a senior developer using **Agile** principles, conducting GitHub PR code reviews and establishing team-wide development standards from scratch
- Responded to a live **production** security incident caused by a compromised npm package, coordinating containment and implementing improved **dependency auditing** practices afterward

PROJECTS

Ant Colony Simulation ([github](#))

Jan 2026

C++, Ubisoft NEXT API

Ubisoft NEXT 2025-2026 Submission

- Solo submission to **Ubisoft NEXT 2025-2026**, simulating two rival ant colonies with pheromone pathfinding and emergent swarm behaviors at **2000 entities** in real-time, built end-to-end in **C++**
- Self-learned and implemented **Entity Component System (ECS)** from scratch with cache-friendly **Structure of Arrays (SoA)** memory layout, **O(1) component lookups** via per-entity bitmasks, and per-type cached entity lists
- Engineered **uniform spatial grid** (32-unit cells) with templated lambda interface for **zero-allocation** neighbor iteration; designed **5-state FSM** for agent AI and a **3-layer pheromone system** producing emergent foraging and combat coordination
- Applied performance optimizations: **sin/cos lookup tables**, **staggered pheromone decay**, **frustum culling**, and **screen-space LOD**

Scriptorium ([github](#))

Oct 2024 - Nov 2024

React, Node.js, Next.js, Tailwind CSS, SQLite, Prisma, Docker, nginx, AWS EC2

CSC309 Course Project

- Built a **collaborative coding platform** supporting writing, executing, and sharing code across **10+ programming languages** as a 3-person CSC309 team project
- Owned **cloud deployment infrastructure**, containerizing the application with **Docker** and configuring **nginx** as a reverse proxy in front of the Node.js backend on **AWS EC2**
- Collaborated on **RESTful API** design with **Node.js**, **Express**, and **Prisma ORM** backed by **SQLite**

Salt, Grooves, and Garlic ([itch.io](#))

Sep 2025 - Present

Unity, C#

CSC404 Game Design Capstone (Team of 10)

- **2nd Best Game Overall** at **LevelUp 2026** showcase, a rhythm-action game built in **Unity (C#)** with a 10-person cross-functional team; architected core game manager and state control systems coordinating scene flow across the full game loop

SKILLS

Languages: C++, C#, JavaScript/TypeScript, Python, Java, C

Backend & Cloud: Node.js, Express.js, RESTful APIs, JWT, RBAC, PostgreSQL, Prisma ORM, Docker, nginx, AWS (EC2), Linux/Unix

Frontend: React, Next.js, Material-UI, Tailwind CSS, Context API

Systems & Practices: ECS, SoA Memory Layout, Spatial Partitioning, FSM Design, OOP, Data Structures & Algorithms, Agile/Scrum, Code Review, AI-Assisted Development (Claude Code, GitHub Copilot)

AWARDS & ACTIVITIES

Awards: Indonesia Maju Scholarship

Teaching: Teaching Assistant for CSC309 (Programming on the Web), University of Toronto

Leadership: Led Recognized Study Group for CSC165 (Mathematical Expression and Reasoning for CS)

Languages: Indonesian (native), English (fluent)